

Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 28]

**The following question explores features of a family of curves. The family is then linked to a homogeneous differential equation.**

Consider the curve given by  $y = \frac{x(x^2 - 16)}{x^2 + 16}$ .

- (a) (i) Sketch the curve of  $y$  for  $-10 \leq x \leq 10$ . [1]
  - (ii) State the coordinates of the points where the curve crosses the  $x$ -axis. [1]
  - (iii) State the coordinates of the local maximum point and the coordinates of the local minimum point. [2]
- (b) State whether the function  $f(x) = \frac{x(x^2 - 16)}{x^2 + 16}$  is odd, even or neither. Justify your answer. [2]

Now consider the general curve given by  $y = \frac{x(x^2 - A)}{x^2 + A}$ , where  $A$  is a positive constant and  $x \in \mathbb{R}$ .

- (c) Given  $f(x) = \frac{x(x^2 - A)}{x^2 + A}$ , prove that  $f'(\sqrt{A})$  is independent of  $A$ . [4]
- (d) (i) Show that  $x - \frac{2Ax}{x^2 + A} \equiv \frac{x(x^2 - A)}{x^2 + A}$ . [2]
- (ii) Hence, determine the equation of the oblique asymptote to the curve. [1]
- (iii) Write down the coordinates of a point on the curve where the oblique asymptote is parallel to the tangent to the curve at that point. [1]

**(This question continues on the following page)**

**(Question 1 continued)**

Now consider the differential equation  $x^2 \frac{dy}{dx} = x(x+y) - y^2$ , where  $x \neq 0$ ,  $y \neq \pm x$ .

Using the substitution  $y = vx$ , the differential equation can be written as  $x \frac{dv}{dx} = 1 - v^2$ .

(e) Using partial fractions, show that  $\int \frac{1}{1-v^2} dv = \frac{1}{2} \ln \left| \frac{A(1+v)}{1-v} \right|$ , where  $A$  is a positive constant. [5]

(f) Hence, show that a solution to the original differential equation may be expressed in the form  $x^2 = \left| \frac{A(x+y)}{x-y} \right|$ , where  $A$  is a positive constant. [5]

Now consider only the case where  $\frac{A(x+y)}{x-y} > 0$ .

(g) Show that a solution to the original differential equation is  $y = \frac{x(x^2 - A)}{x^2 + A}$ . [4]